MILITARY SPECIFICATION

HONING MACHINE, CYLINDER RESIZING

This specification is approved for use by the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a floor-mounted, automatic, honing machine for resizing engine cylinders.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

CC-M-1807	- Motors, Alternating Current, Fractional and Integral Horsepower (500 HP and Smaller).
PPP-D-729	- Drums, Shipping and Storage, Steel, 55-Gallon (208 Liters).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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MIL-T-152	- Treatment, Moisture and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment.
MIL-V-173	- Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).
MIL-M-18058	- Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

US DEPARTMENT OF LABOR

OSHA 2206	- General Industry, OSHA Safety and Health Standards (29 CFR 1910).

DRAWINGS

US ARMY

7346610	- Cylinder Assembly

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02	- Manual for Machine Tool Gearing.
AGMA 390.03	- Gear Handbook, Volume I, Gear Classification, Materials, and Measuring Methods for Unassembled Gears.

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(Application for copies should be addressed to the American Gear Manufacturers' Association, 1901 N. Fort Myer Dr., Arlington, VA 22209.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametral Pitches.
ISO 1122/1 - Glossary of Gear Terms - Part 1: Geometrical Definitions.

(Application for copies of ISO documents should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.
ASTM E380 - Standard for Metric Practice.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - General Standards for Industrial Control and Systems

(Applications for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, NW, Washington, DC 20037.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 - National Electrical Code.
NFPA 79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery.

(Application for copies should be addressed to the National Fire Protection Association, Battery March Park, Quincy MA 02269.)

RUBBER MANUFACTURERS' ASSOCIATION (RMA)

IP-23 - Single V-Belts (2L, 3L, 4L, and 5L Cross Sections).

(Application for copies should be addressed to the Rubber Manufacturers' Association, 1400 K Street, NW, Washington, DC 20005.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J431 - Automotive Gray Iron Castings.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

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(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more complete honing machines for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

3.2 Design. The honing machine shall consist of a base and housing, spindle, drive tube, electric motor drive, and any associated honing heads, gages, and feeds required for operation. The honing machine shall be capable of honing cylinder bores in 60 degree and 90 degree V-type blocks, in-line blocks, and angle head blocks up to 42 inches in length. It shall also be capable of honing individual thin walled, air cooled engine cylinders and engine cylinders conforming to Drawing 7346610. The machine shall have a single spindle for resizing (honing) cylinder bores from two inches to at least 6.25 inches in diameter, and 4.5 inches to 11 inches in cylinder bore length.

* 3.2.1 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.2 Safety and health requirements. Safety devices shall be provided for all parts presenting safety hazards. Guards shall be removable to provide easy access to guarded parts. The honing machine shall comply with the general safety and health requirements promulgated under OSHA 2206 that are applicable to the machine itself. Exceptions and additional requirements for safety and health shall be as specified in the contract (see 6.2.1 and 6.4).

* 3.2.3 Lubrication. All bearings (except sealed-for-life type), mating gears, and sliding parts shall be provided with standard means for lubrication. Oil holes, grease fittings, and filler caps shall be accessible without requiring disassembly of functional parts. Lubrication reservoirs, if applicable, shall have means for checking levels.

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* 3.2.4 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.2.4.1 Reclaimed materials. The manufacturer is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.3 Construction. The honing machine shall be constructed to be capable of performing the operations specified herein. The machine and its equipment shall be complete so that, when installed and connected to a proper power source, it can be used for all operations for which it was designed.

* 3.3.1 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

* 3.3.2 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

* 3.3.3 Fastening devices. All screws, pins, bolts, and similar parts shall be installed with means for preventing loss of tightness and adjustment. The methods for preventing loss of tightness or adjustment shall be in accordance with accepted engineering standards and practice. All such parts when subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently deformed.

3.3.4 Surfaces. All castings, forgings, molded parts, stampings, and welded parts shall be clean and free from sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials. External surfaces shall be smooth and all edges shall be either rounded or beveled.

* 3.3.5 Painting. Unless otherwise specified (see 6.2.1), painting and finishing shall be in accordance with the best commercial practice in the industry. Before painting, all surfaces shall be clean and be free of all foreign matter detrimental to painting. At least one coat of primer (or a phosphate or chromate base) and one coat of enamel, or equivalent (i.e., epoxy enamel), shall be applied.

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* 3.3.6 Gears. All gears and pinion designs selected for use in the honing machine or its components shall be fabricated to either English (USA) standard inch diametral pitch per AGMA 360.02 and AGMA 390.03 or Metric (SI) module ratio per the International Organization for Standardization (ISO) ISO 54 and ISO 1122 and ASTM E380.

3.3.7 Belts and pulleys. Belt drives shall be in accordance with RMA IP-23. Pulleys shall be made of cast iron or aluminum, and shall be balanced. A belt guard shall be provided to protect the operator during all machining operations (see 3.2.2).

* 3.3.8 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.3.9 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

* 3.4 Performance requirements. The machine shall be capable of stock removal and finish honing as follows:

(a) Any continuous, plain, open end bore to within ±0.0002 inch of desired diameter and to a true cylindrical surface within ±0.0005 inch total indicator reading, with a finish of 8-16 microinches arithmetical average.
(b) Any tapered and out-of-round bore within the established capacity.
(c) Maintain the profile and finish as required on Drawing 7346610, while resizing a Continental 1790 cylinder to within its original rebuild limitations of 0.008 inch maximum oversize from standard in not more than 15 minutes, including set-up time.
(d) Maintain the profile and finish as required on Drawing 7346610, while resizing a Continental 1790 cylinder to 0.018 inch oversize to prepare the cylinder for metalizing in not more than 25 minutes, including set-up time.
(e) Maintain the profile and finish as required on Drawing 7346610, while resizing a moly-coated Continental 1790 cylinder back to standard size, removing approximately 0.015 inch moly metal in not more than 25 minutes, including set-up time.
(f) Resize both banks of a standard commercial V-8 engine block to 0.020 inch oversize in accordance with (a) above in not more than 50 minutes, including set-up time.
(g) Produce the cylinder wall cross hatch or scratch pattern, as specified on Drawing 7346610. The honed finish of the cylinder bore shall be analyzed by performing transparent impressions of the finished hone pattern at locations opposite the flange and the mid bore and opposite the threaded section. The transparent impression shall be magnified to a minimum of 100X and analyzed for the following:
    Cross hatch angle
    Cross hatch angle to be cut uniformly in both directions
    Cross hatch angle to be clean cut, but not sharp, and free from torn and folded metal
    The plateau area shall be one-half (1/2) to two-thirds (2/3) of the surface area

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The plateau area shall be free from a burnished or glazed surface
The surface shall be free of imbedded particles

3.5 Components.

3.5.1 Base. The base shall be a heavy, rigid, impact resistant structure designed for floor mounting. The base shall be constructed of cast iron or iron alloy, or shall be fabricated from structural steel. Sectional construction of two or more sections shall, at final assembly, offer the same rigidity and accuracy as a single structure. The structure shall be internally ribbed and braced to minimize distortion and deflection. Doors or removable covers shall be provided to permit servicing the drive motor and other internal mechanisms. Guides and a traverse mechanism with a handwheel shall be provided to move the drive head assembly sideways from cylinder to cylinder. The base shall support the splash guards, which enclose the work area to prevent loss of honing oil or coolant. One side of the splash guards shall be arranged to allow complete access to the work area for loading and unloading the honing machine.

3.5.2 Drive head assembly. The drive head assembly shall be constructed of impact resistant metal. The assembly shall house the spindle drive mechanism, spindle, stone feed control, stock removal dial, manual stone contraction control, stroke adjustment control, quick start-and-stop lever, and a micrometer feed with a positive setting to prevent "creep" in the cylinder. The internal reciprocating action of the drive head shall be mechanical or hydraulic, and designed to properly reciprocate within the range of speeds covered by this machine. The drive head assembly shall be rigidly supported on the base and shall be guided by steel bars, bedways, or other suitable means.

3.5.3 Spindle. The spindle shall be of alloy steel, mounted in antifriction ball or roller bearings, and designed to receive a honing head. The spindle rotary motion shall be obtained through a belt drive, sliding gears, or other equally positive and suitable means. Adjustment and operation of reciprocating motion shall be independent of the rotary motion and so arranged that, during normal operation, the reciprocation and rotation of the spindle are simultaneous. The spindle drive mechanism shall have a mechanical quick start-and-stop mechanism and shall work in conjunction with the abrasive stone feed-up-control to automatically expand the abrasive stones and stop the machine when the predetermined limits of honing have been reached.

3.5.4 Stroke adjustment control. Stroke adjustment control shall provide adjustment of stroke length for cylinders 4.5 inches to 11 inches in length. The stroke length scale shall be graduated, in the inch system, in increments not more than 0.250 inch. Stroke length scale(s) shall be calibrated for each stone length furnished with the honing machine. Stroke adjustment control shall automatically reverse the travel of the drive assembly at the extreme top and bottom of each stroke. A stroke position indicator light shall indicate when the honing head is at the bottom of the stroke.

3.5.5 Honing head. The honing machine shall have a free-floating, self-aligning honing head that can follow the axis of the existing bore of the workpiece. The honing head shall use replaceable honing stones and shall have provisions for automatic incremental stone out-feed while the head is being reciprocated and rotated. Honing heads and stones capable of honing the bore diameter, stroke lengths, and capable of producing the metal removal rates and surface finishes specified herein shall be furnished with each machine.

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* 3.5.6 Coolant system. The coolant system shall include an enclosed sump or reservoir, a power driven recirculating pump, oil cooler, blower, and piping. The system shall incorporate a strainer, which is removable for cleaning, to remove cutting chips from the coolant. The recirculating coolant system shall include oil resistant hoses, tubing, or piping to direct the coolant. The system shall include a delivery nozzle and valve to permit the operator to direct and control the amount of coolant flow. The capacity of the coolant reservoir shall be not less than 35 gallons. There shall be a honing oil filter unit connected to the basic coolant system. That filter unit may be either external (see 3.5.6.1) to, or an integral part of, the honing machine. The unit shall have not less than the following minimum characteristics:

(a) Five micron filtration.
(b) Ten gallon per minute flow capacity at 25 pounds per square inch.
(c) 9100 square inches of filter area.
(d) 150 cubic inches of sludge capacity.
(e) Seven gallon filter container oil capacity.
(f) One horsepower pump motor.
(g) Initial filter element(s) and additional elements for three filter element replacements.
(h) The filter unit shall operate from 230 volts, three phase, 60 hertz (Hz), with a separate "OFF-ON" switch and an indicator "ON" light.

* 3.5.6.1 External coolant filter. If an external filter unit is provided, it shall consist of a steel cabinet of not less than Manufacturers Standard Gage No. 10. The cabinet shall be not larger than 30 inches wide, 23 inches deep, and 33 inches high, with top and front service doors.

* 3.5.7 Plates. All speed and feed indicating plates and instruction plates shall be marked in the English system. Markings shall be permanent and shall be in plain bold-faced characters, on a contrasting background, that can be easily read.

3.5.8 Controls. All operating and adjustment controls, meters, hand wheels, levers, etc., shall be located at the front of the machine within the operator's station. The machine shall be convenient to operate by one operator, with all controls within easy reach.

3.5.9 Stone feed control. A stone feed control shall be provided on the drive head assembly and shall contain a micrometer feed dial to facilitate in setting the stone for the desired stock removal. The dial shall be graduated in increments of 0.001 inch. It shall be possible to adjust the stone feed control for additional stock removal while the honing head is being reciprocated and rotated in the cylinder.

3.5.9.1 Dwell mechanism. The machine shall have an electric, hydraulic, or combination thereof dwell mechanism that shall cause the honing head to dwell or hesitate at the bottom of the stroke to facilitate removal of cylinder taper. The time period of dwell shall be adjustable, from zero to not less than three seconds, by means of a calibrated dwell control. When actuated, the dwell mechanism shall automatically dwell the honing head at the bottom of the cylinder within the predetermined time period selected.

3.5.9.2 Load meter. A load meter shall be provided on the machine for indicating stone cutting action and cylinder straightness. The load meter shall be calibrated to indicate a cylinder straightness of 0.001 inch per six inch length.

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* 3.5.10 Electrical equipment. Unless otherwise specified (see 6.2.1), all electrical equipment shall conform to NEMA ICS 1 and NFPA 79, as applicable. The machine shall be powered by a nominal 230/460 volt, 60 Hz, three phase electrical power source. Unless otherwise specified (see 6.2.1), the machine shall be initially wired to operate from a 208/230 volt, 60 Hz, three phase power source. Control circuits shall operate from 115 volt single phase power.

* 3.5.11 Motors. Motors shall conform to the applicable requirements of CC-M-1807, with Grade B insulation or better. The machine shall be equipped with the following minimum motor sizes: a two horsepower (hp), 60 Hz, three phase drive motor; a 1/4 hp pump motor for the oil cooling system; and a 1/2 hp blower motor.

* 3.5.12 Electric power supply cable. An electrical power supply cable shall be installed on each machine. The cable shall be not less than ten feet in length and shall terminate in not less than one inch of stripped tinned wire, without a receptacle plug. The cable shall include proper sheathing/covering in accordance with NFPA 70. The cable shall have a voltage/potential and ampere/current carrying capacity exceeding that required by the electric power consuming devices, including electric motors, of the honing machine in accordance with NFPA 70.

3.6 Equipment.

3.6.1 Standard equipment. The following equipment shall be furnished with each machine as standard equipment:

(a) Clamps that shall align and hold the block in a rigid position.
(b) A micrometer feed with a positive setting to prevent "creep" in the cylinder to be honed.
(c) Gages for setting honing heads, guides, and honing stones for the proper diameter.
(d) An indexing quadrant and locating lever for positioning V-type blocks.
(e) Fixtures for inserting honing stones in the master holder sets that are used in the honing head.

3.6.2 Additional equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

(a) One set of wrenches normally furnished with the machine.
(b) One each of the heavy-duty honing heads necessary for resizing (honing) cylinder bores from two inches to at least 6.25 inches in diameter (see 3.2). Each honing head supplied shall be provided with alignment guides, master holder sets, and guide shoe sets for roughing and finishing stones.
(c) One honing head for honing cylinders conforming to Drawing 7346610 with alignment guides, master holder sets, and guide shoe sets for roughing and finishing stones.
(d) One fixture for holding the cylinders conforming to Drawing 7346610, with clamps and necessary alignment guides.
(e) Twelve sets (roughing and finishing) of honing stones for each head supplied.
(f) One stone inserter fixture for inserting honing stones in each of the master holder sets of the honing heads supplied, as applicable.

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(g) One each setting gage for setting honing stones, guides, and head to properly align and set the diameter for the honing heads supplied, as applicable.
(h) One adjustable work light.
(i) Fifty-five gallons of standard honing oil.
(j) Fifty-five gallons of special honing oil for use in honing the air cooled cylinders specified on Drawing 7346610.

3.6.3 Optional equipment. Optional equipment shall be as specified in the contract or order (see 6.2.1).

3.7 Lubrication charts or plates. Lubrication charts or plates shall be protected by being laminated between clear plastic sheets or by being permanently inscribed on metal. The charts or plates shall be secured to the machine and shall include the following information:

Points of application
Service interval
Type of lubricant
Vicosity
Federal or Military specification number for the lubricant, when applicable.

* 3.8 Nameplate. The nameplate and information plates may be incorporated into one or more corrosion-resistant metallic plates at the option of the manufacturer. The nameplate shall have permanently legible characters, printed in the English language, identifying the honing machine specified herein. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The nameplate shall be secured with permanent fasteners at each of the four corners. The following information shall be included on the nameplate:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number
Power Input (volts, amps, phase, frequency/Hz, alternating current)
Horsepower

3.9 Instruction panel. An instruction panel, describing operating and service procedures, shall be prominently located, and permanently attached to, the honing machine.

* 3.10 Workmanship. Standards of workmanship shall assure that the honing machine shall have the stability, strength, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (d) as follows. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.5).
(b) Operational test (see 4.6.1).
(c) Accuracy test (see 4.6.2).
(d) Packaging inspection (see 4.7).

* 4.4.1 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.

* 4.5 Examination. Visually and manually examine the honing machine to determine conformance with the requirements of 3.2 through 3.2.4.1, 3.3 through 3.3.9, and 3.5 through 3.10. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Manual examinations shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the machine and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Tests. All tests shall be conducted in accordance with the contractor's instruction manual supplied with the machine (see 6.3). The honing oil filter unit of 3.5.6 shall be in operation for all tests, and shall be inspected and tested to verify compliance with specified requirements. All filters used during test shall be replaced with new, unused filters for delivery on the contract. All applicable equipment in 3.6.1 and 3.6.2 furnished with the machine shall be used

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during tests. All honing stones and honing oil used during tests shall also be replaced for delivery. Following tests, the machine shall be cleaned and prepared for delivery.

4.6.1 Operational test. An operational test shall be conducted on the machine at no load to assure proper operation of all components, controls, and mechanisms to verify compliance with all functional requirements specified in Section 3 and the contract.

4.6.2 Accuracy test. Two open end test cylinders, one of the smallest diameter and length and the other of the largest diameter and length specified in 3.2, shall be set up and honed their entire length. The test cylinders shall be automotive gray iron castings in accordance with SAE J431, grade 4000, 217-269 Bhn (4.1 - 3.7 Brinell impression diameter). A minimum of 0.020 inch of metal shall be removed and the cylinders inspected to verify compliance with the accuracy and surface finish requirements specified in 3.4(a).

4.6.3 Cylinder assembly honing tests.

(a) One cylinder assembly shall be honed to the profile depicted on Drawing 7346610 within its original rebuild limitation of 0.008 inch maximum oversize from standard in not more than 15 minutes, including set-up time. Inside diameters, tapers, surface finish, and scratch pattern shall be as specified on the drawing (see 3.4(c)).

(b) One cylinder assembly shall be honed to the profile depicted on Drawing 7346610 to 0.018 inch oversize in not more than 25 minutes, including set-up time. Inside diameters, tapers, surface finish, and scratch pattern shall be as specified on the drawing (see 3.4(d)).

(c) One metalized cylinder assembly shall be honed to the profile depicted on Drawing 7346610, resizing back to standard size, in not more than 25 minutes, including set-up time. Inside diameters, tapers, surface finish, and scratch pattern shall be as specified on the drawing (see 3.4(e)).

(d) One commercial V-8 engine block shall be honed to 0.020 oversize in not more than 50 minutes, including set-up time (see 3.4(f)).

* 4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-M-18058.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. The machine covered by this specification is intended for use in resizing and finish honing standard size engine block cylinders, for blocks to 42 inches in length, individual thin walled air cooled engine cylinders, and engine cylinders conforming to Drawing 7346610.

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6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title number, and date of this specification.
(b) First article when required (see 3.1).
(c) Exceptions and additional safety/health requirements when required (see 3.2.2 and 6.4).
(d) Painting and finishing if different (see 3.3.5).
(e) Electrical equipment if different (see 3.5.10).
(f) Voltage requirements for initial wiring if different (see 3.5.10).
(g) Additional equipment if different (see 3.6.2).
(h) Optional equipment if any (see 3.6.3).
(i) First article inspection if different (see 4.3).
(j) Quality conformance inspection if different (see 4.4).
(k) Sampling when required (see 4.4.1).
(l) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data, such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract (see 4.6 and 6.3.1).

6.3.1 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturer (OEM - actual part manufacturer),, Federal Supply Code for Manufacturers (FSCM), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

* 6.4 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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